

THE RISE OF THE MACHINES

Welcome to Rise of the Machines, Enova's flagship robotics proving ground where raw mechanics, embedded systems, and autonomous logic collide on the battlefield! This category pushes your robot building limits through rapid circuit soldering, pin assignment optimization, sensor calibration trials, and high-stakes autonomous navigation challenges. Victory demands structural durability, and relentless adaptability under pressure, so power up your robot and command your machine to master the floor!

Round 1 : The World Reimagined

ROUND DURATION: 1.5 Hours

DELEGATE CAP: 3

Fast forward. Happy New Year! The Earth in 2081 is to be fully dependent upon Machines. Your task is to make a presentation of 5-8 specialized robots which help our planet to be more efficient. You can include robots which replace heavy human labor with coordinated cranes, massive 3D printers, and brick-laying robots to build skyscrapers. Robots that lend a hand in household tasks, take over hospitals as doctors or nurses. Machines at supermarkets instead of humans, security robots etc.

Keep in mind that your presentation should include:

- 3D, preferably, or 2D simulations of each robot
- The name and type of the robot
- Its dimensions and features
- The purpose of the robot
- What materials it is made from
- Why it is made from these materials
- How it benefits humans
- How it benefits the environment
- Its limitations (if any)

Note:

- Each one of these pointers should be addressed in detail for each robots.
- This presentation should be emailed to triathlon.opf8@gmail.com by Wednesday, 16th September, 11:59 P.M.

Judging Criteria for Round 1

The presentations will be judged on the level of details of the robot (dimensions and features), 3D/2D models of the robots, its cost effectiveness, benefits to humans and environment and the diverse nature of robots present. Note that each of these points will be individually applied to each robot in the presentation .

Round 2 : Buzzer Blitz

ROUND DURATION: 35 Minutes

DELEGATE CAP: 2

This round is an engaging and knowledge-driven segment designed to test participants' understanding of robotics, automation, electronics, and emerging technologies. The round will challenge students with a variety of conceptual, technical, and scenario-based questions that require logical thinking and problem-solving skills. Through this round, participants will demonstrate not only their theoretical knowledge but also their ability to apply robotics concepts to real-world situations.

Important:

- Make sure to take a look at the study guide thoroughly before Round 2!

Guide:

- https://docs.google.com/document/d/1nfxZx8FI0k_4YpQYsrWxR1BIG9fBGrhuw4eOXsqRw5Q/edit?usp=drivesdk

Judging Criteria for Round 2

This round features a 20 mark quiz which and to qualify for Round 3 the team must score a minimum of 50% (10 marks).

Round 3 : Crisis Mode

ROUND DURATION: 2.5 Hours

DELEGATE CAP: 3

- 1) Earth faces growing environmental threats, from devastating wildfires and sudden flash floods to collapsing ecosystems and endangered wildlife. In this round, you will take on the role of a robotics engineer and design a specialised robot capable of tackling a real-world environmental crisis, protecting ecosystems or assisting communities in danger. The problem is real. The solution is yours. What will you build?
- 2) Space exploration faces growing challenges, from meteoroid showers and radiation to extreme temperatures and unstable terrains. Plus human interest in exploring the unknown features of space is heavily increasing. In this round, you will take on the role of a robotics engineer and design a specialised robot capable of tackling a real-world space crisis, adapting to its environment and helping humans explore or operate safely. The problem is real. The solution is yours. What will you build?
- 3) Modern transportation faces growing challenges, from traffic congestion and delayed supply chains to dangerous transport conditions and rising carbon emissions. In this round, you will take on the role of a robotics engineer and design a specialised robot capable of tackling a real-world transportation crisis, whether navigating cargo, supporting emergency transit, or safely transporting essential goods such as organic food and chemicals. The problem is real. The solution is yours. What will you build?

Remember your robot should be:

- Able to fulfill the purpose it was designed for.
- Able to fit the specific features designated.
- Able to carry out the primary task it was given.
- You should be able to pitch your idea to the judges.

Important:

- Delegates can not choose their own topic from the above stated topics rather they will be assigned any one of these topics on the day of Round 3. Teams will be provided with cardboard and a limited set of sensors: ultrasonic, infrared, temperature and soil sensors. Only the designated sensors will be provided by the organisers. All other equipment, components and materials required to construct and operate your robot must be brought by the participating teams themselves. Teams should therefore come prepared with equipment that can be appropriately integrated with the provided sensors and used to build their proposed solution. Your preparation should focus on robotics, sensors, programming and problem-solving. Consider your robot's environment, design, size, weight, efficiency and safety. There is no single correct solution—use your resources wisely and turn your idea into a functional robot.

Safety Notice:

- Contestants must not bring any prohibited, hazardous, flammable, or unsafe items to the competition. All equipment brought by teams will be subject to safety checks and approval by the organiser

Judging Criteria for Round 3

For this round the robot you made will be judged on the basis of how effectively it is able to fulfil the purpose it was designed for, how its features align with its work, how efficiently teams have used the provided equipment combined with what things they brought themselves and how they're able to present their robot and pitch its idea to the judges.